Heat as witness of quantum properties

A. de Oliveira Junior

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Technical University of Denmark



Outlook

I. Statement of the problem

II. Optimal heat flow

III. Heat-based witnesses

In collaboration with



Jonatan Bohr Brask

Technical university of Denmark



Patryk Lipka Bartosik

Center for Theoretical Physics

Based on:

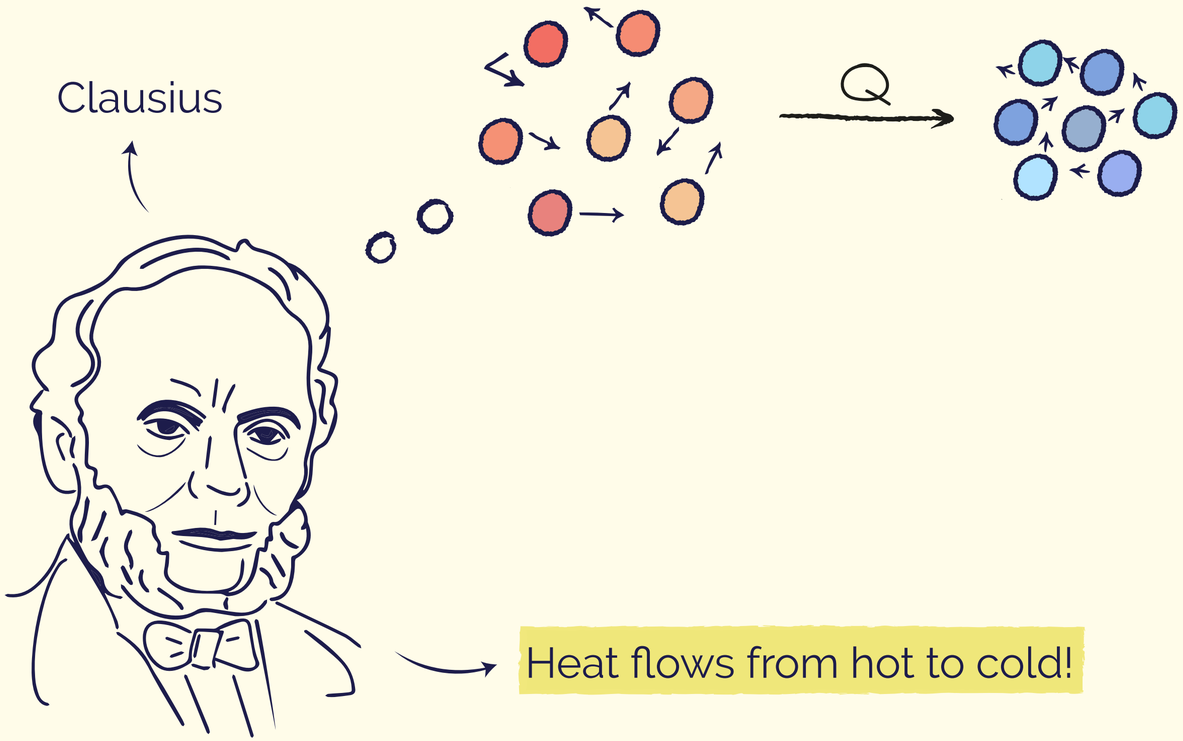
Phys. Rev. Lett. 134, 050401 (2025)
[arXiv: 2408.06418]

I. Statement of the problem

How do *quantum features* affect thermodynamic processes?



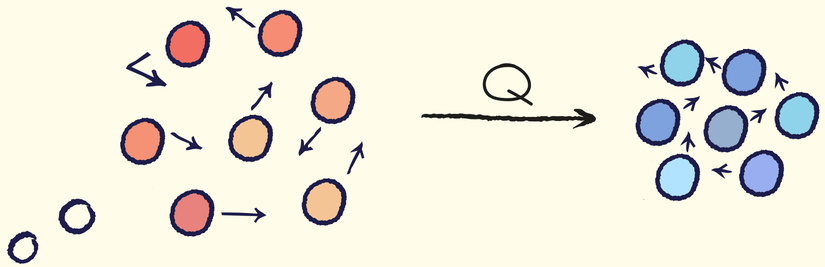
How do *quantum features* affect thermodynamic processes?



Heat flows from hot to cold!

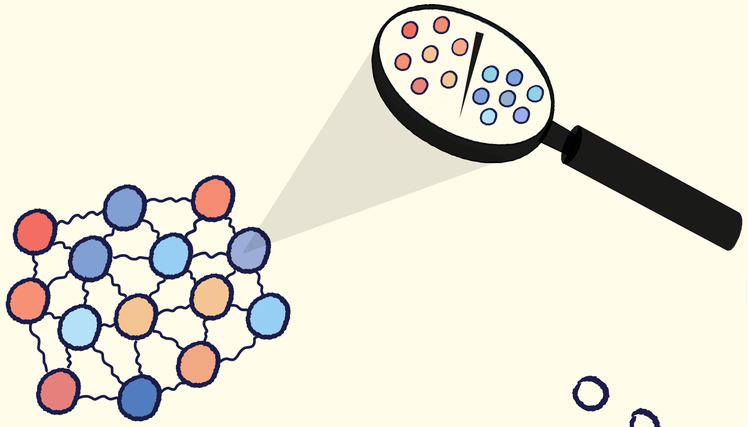
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Clausius





Heat flows from hot to cold!



Devilish Clausius



Heat can flow from cold to hot!

M. H . Partovi, *Phys. Rev. E* ***77****, 021110 (2008)*

D. Jennings & T. Rudolph, *Phys. Rev. E* ***81****, 061130 (2010)*

Really ***quantum***?

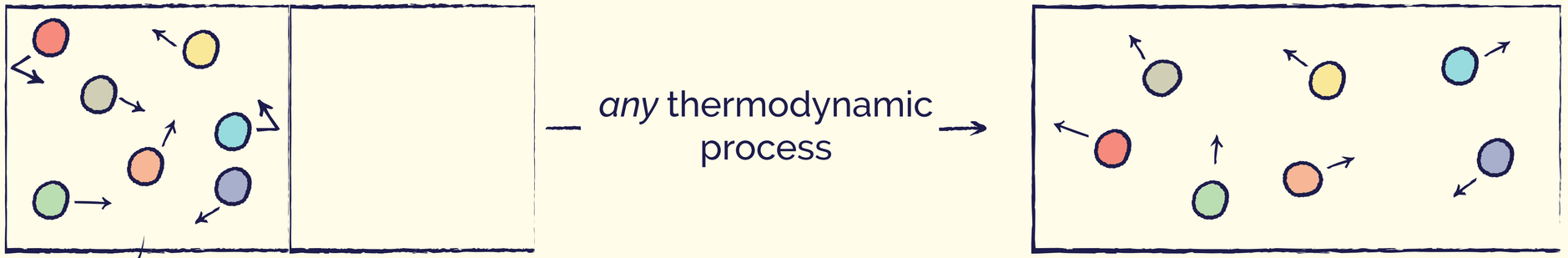


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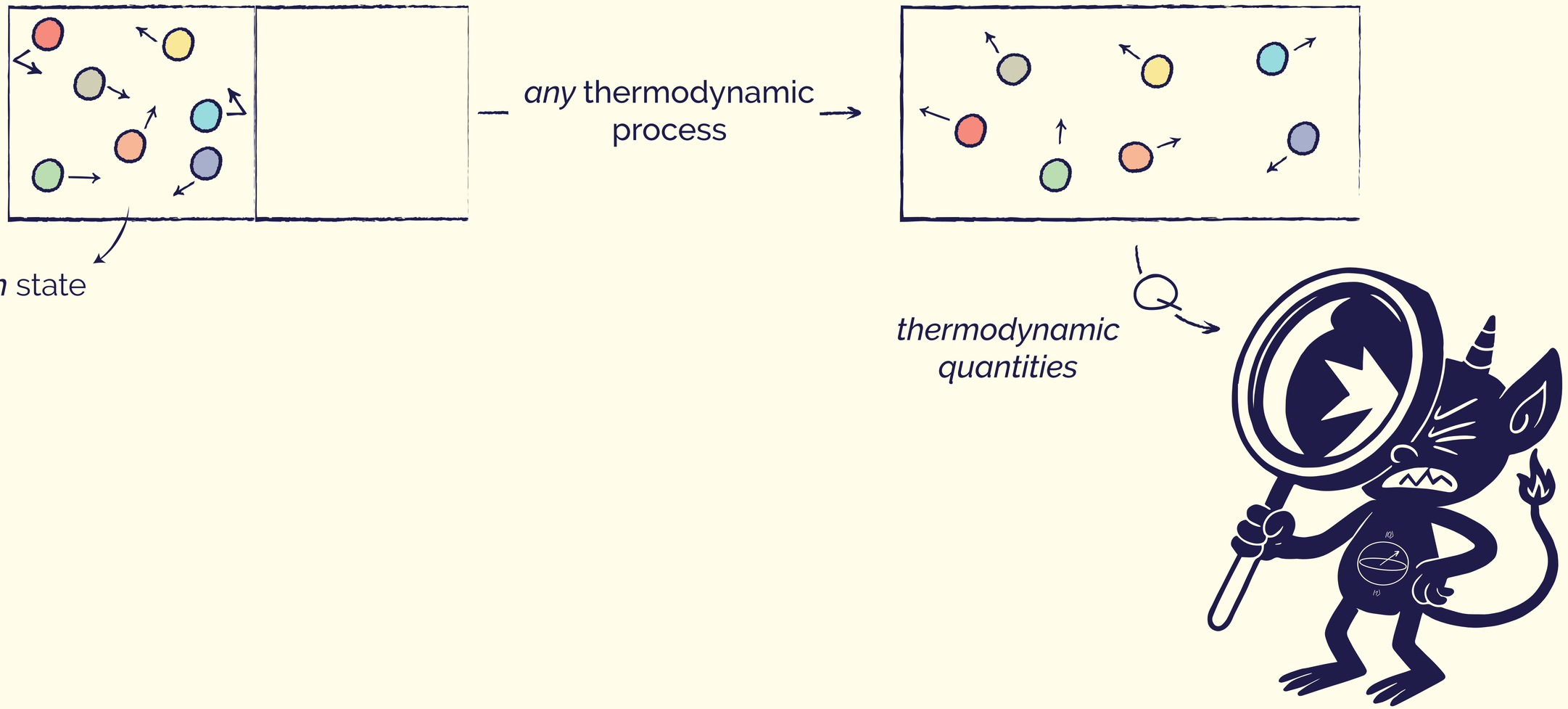
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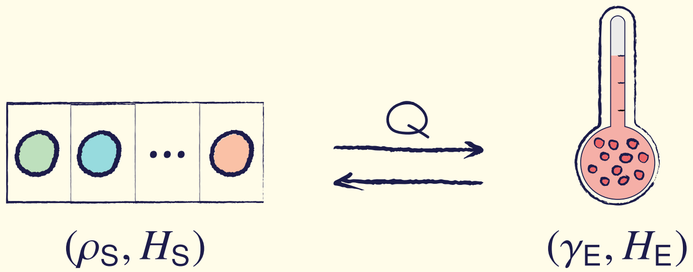


...but how?

II. Optimal heat flow

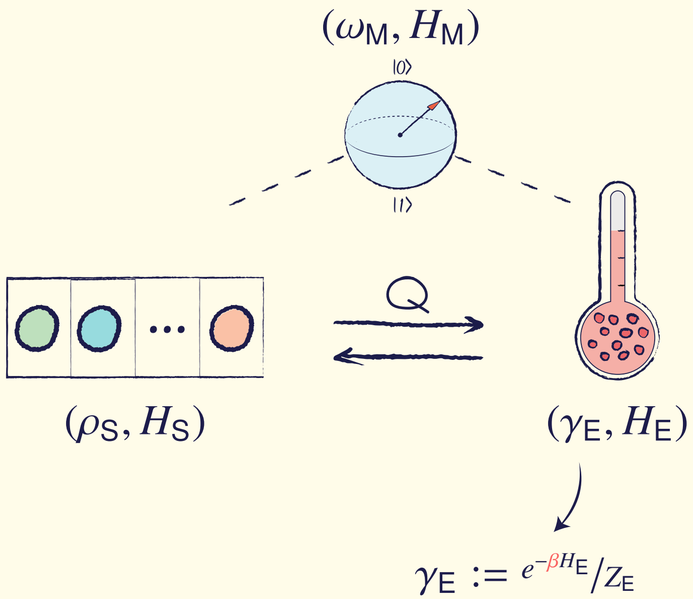
Scenario

- Composite system:



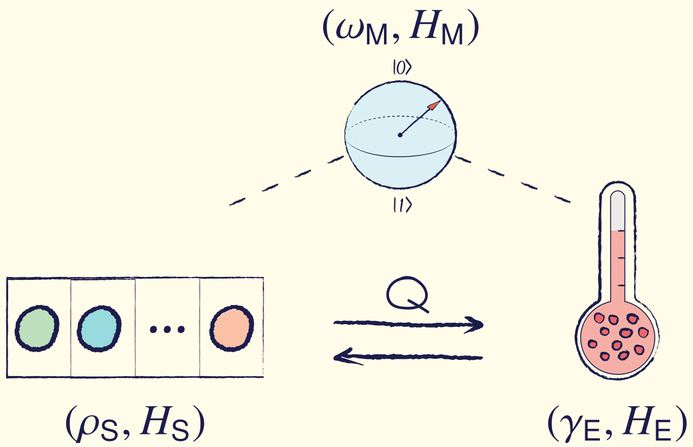
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No assumptions about

- Interaction's strength, complexity, duration
- Only heat exchange

i. SME evolves under an energy-preserving unitary:

$$\sigma_{SME} := U(\rho_S \otimes \omega_M \otimes \gamma_E)U^\dagger,$$

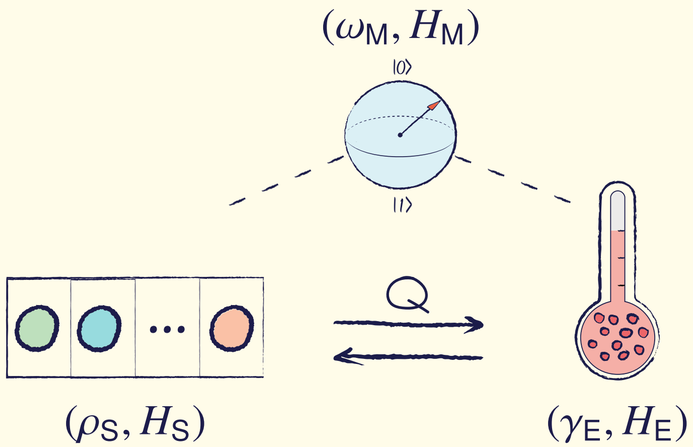
where $[U, H_S + H_M + H_E = 0]$.

P. Lipka-Bartosik, G.F Diotallevi, P. Bakhshinezhad, *Phys. Rev. Lett.* ***132***, 140402 (2024)

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No heat from M **to** SE

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$$\sigma_M := \mathrm{tr}_{SE}\, U(\rho_S \otimes \omega_M \otimes \gamma_E)U^\dagger = \omega_M.$$

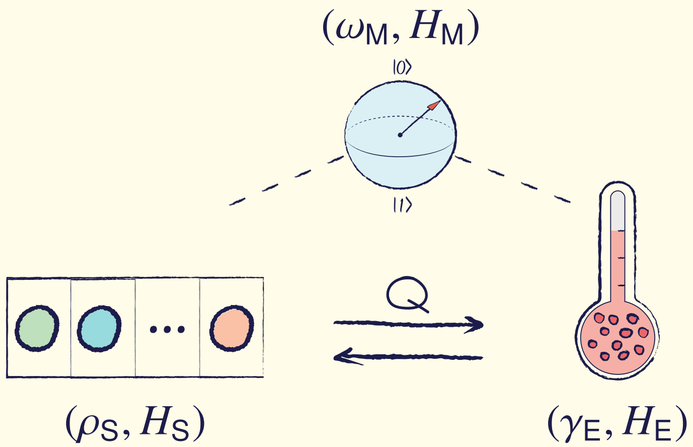
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No complicated way to infer Q

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ii. The state of the memory is fixed by the dynamics:

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iii. Probe the heat exchange between SE:

$$Q_E(\beta, H_E, U, \omega_M, H_M) := \mathrm{tr}[H_E(\sigma_E - \gamma_E)],$$

where $\sigma_E := \mathrm{tr}_{SM}(\sigma_{SME})$.

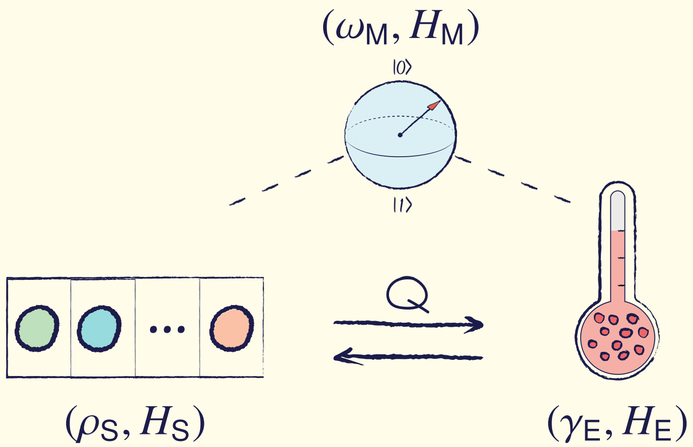
What's the **optimal** heat flow?

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$$Q_{c/h}(\rho_{\mathsf{S}}) := \min_{H_{\mathsf{E}}, H_{\mathsf{M}}, U, \rho_{\mathsf{M}}} / \max \ \mathrm{tr}[H_{\mathsf{E}}(\eta_{\mathsf{E}} - \rho_{\mathsf{E}})],$$
$$\text{s.t.} \quad [U, H_{\mathsf{S}} + H_{\mathsf{M}} + H_{\mathsf{E}}] = 0,$$
$$\eta_{\mathsf{M}} = \rho_{\mathsf{M}}.$$

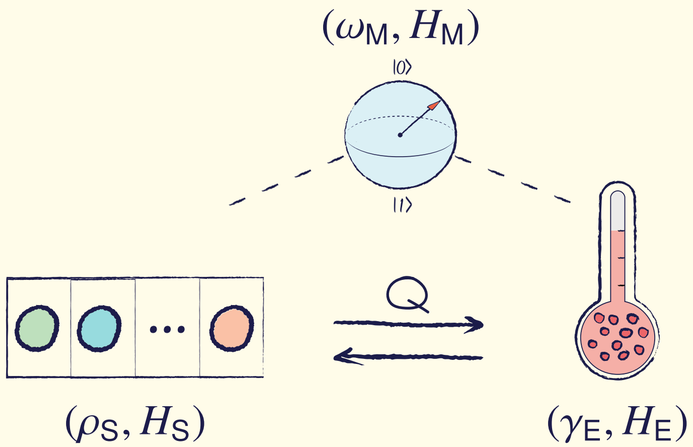
very nasty!

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What's the **optimal** heat flow?

$$Q_{c/h}(\rho_S) := \min_{\eta_S}/\max \ \mathrm{tr}[H_S(\rho_S - \eta_S)],$$
$$\text{s.t.} \quad F_\beta(\rho_S) \geq F_\beta(\eta_S)$$

Still nasty, but solvable!

$$F_\beta(\rho) := \mathrm{tr}(\rho H) - \beta^{-1} S(\rho)$$

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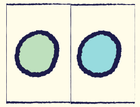
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II. *Heat-based* witnesses

Optimal heat exchange

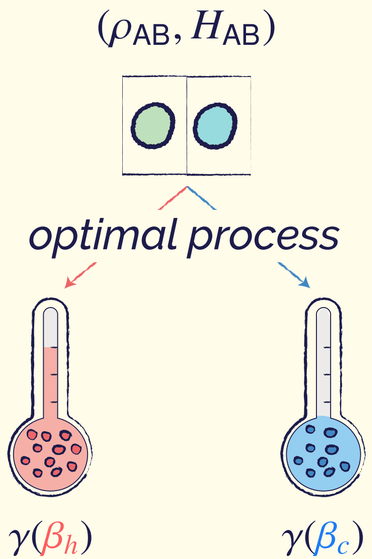
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$$(\rho_{\mathrm{AB}}, H_{\mathrm{AB}})$$



Optimal heat exchange

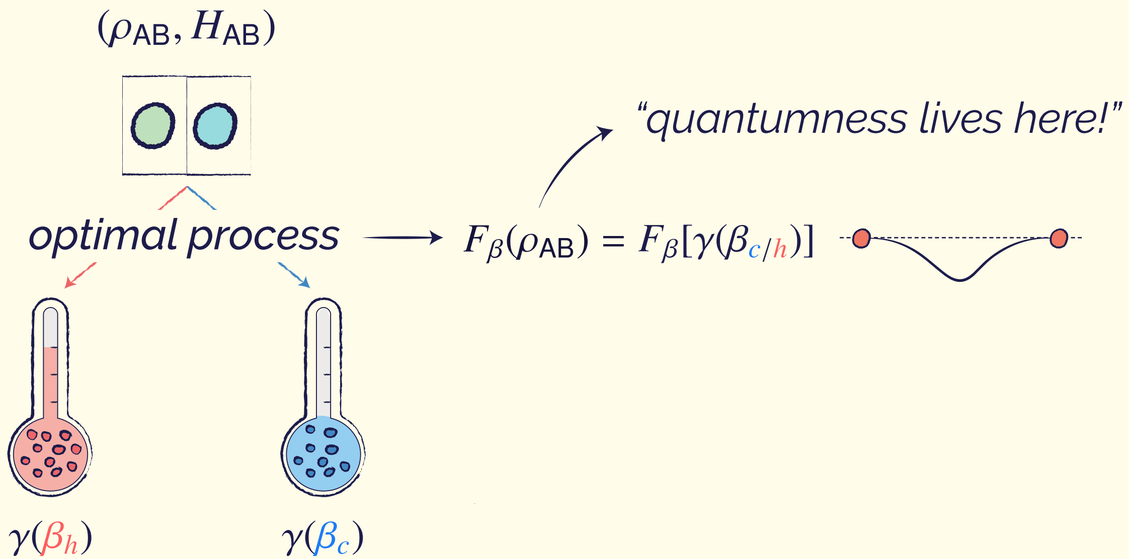
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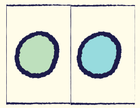


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optimal process $\longrightarrow$ $F_\beta(\rho_{\mathrm{AB}}) = F_\beta[\gamma(\beta_{c/h})]$  $\longrightarrow$ $Q_{c/h}(\rho_{\mathrm{AB}}) = Q^{\star}_{c/h}(\mathcal{S}_{\mathrm{SEP}})$ + *quantum contribution*

same E_{AB} & marginal entropy as ρ_{AB}



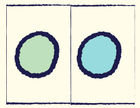
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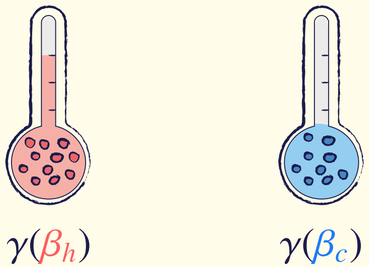
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Conditional entropy in a nutshel

- $S(\mathrm{A}|\mathrm{B})_{\rho_{\mathrm{AB}}} := S(\rho_{\mathrm{AB}}) - S(\rho_{\mathrm{B}})$
- $S(\mathrm{A}|\mathrm{B})_{\rho_{\mathrm{AB}}} < 0 \Longrightarrow \rho_{\mathrm{AB}}$ is entangled!

N. Cerf & C. Adami, Phys. Rev. Lett. 79, 5194 (1997)

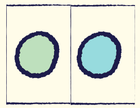
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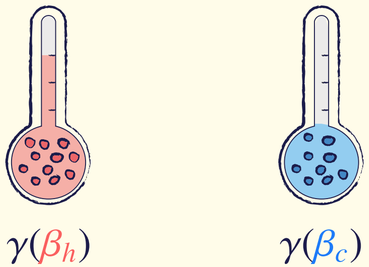
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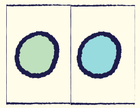
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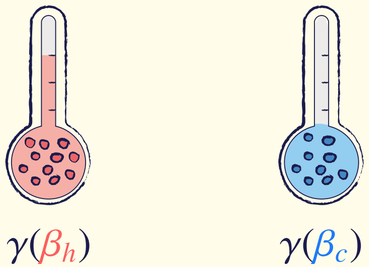
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! Our results are *general*:

- *Optimal* heat flow for $\mathrm{S} = \mathrm{A}_1 \ldots \mathrm{A}_N$
- *Other* properties (coherence, magic, etc)

HIERARCHY!



States with the same E_{AB} & $S_{\mathrm{A}}/S_{\mathrm{B}}$:



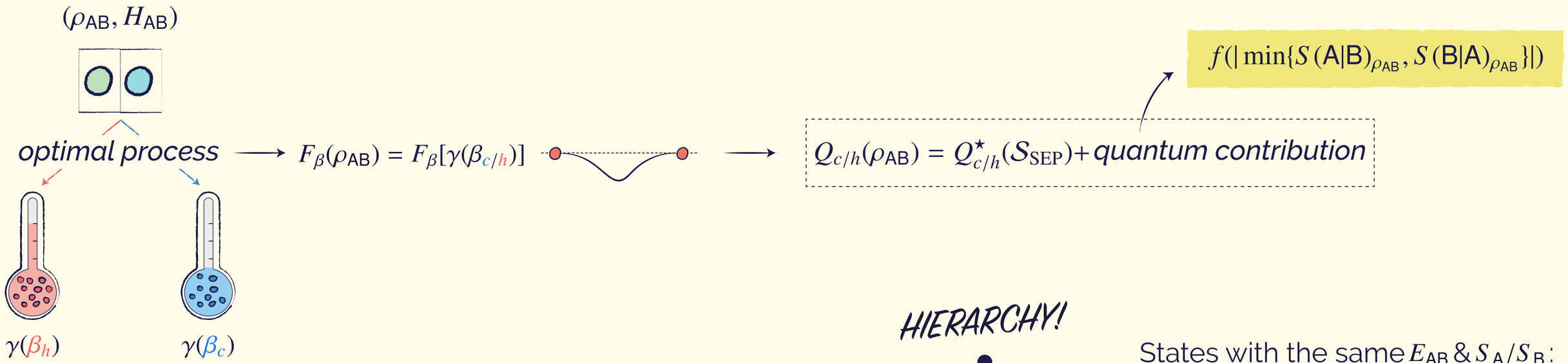
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heat-based witnesses?

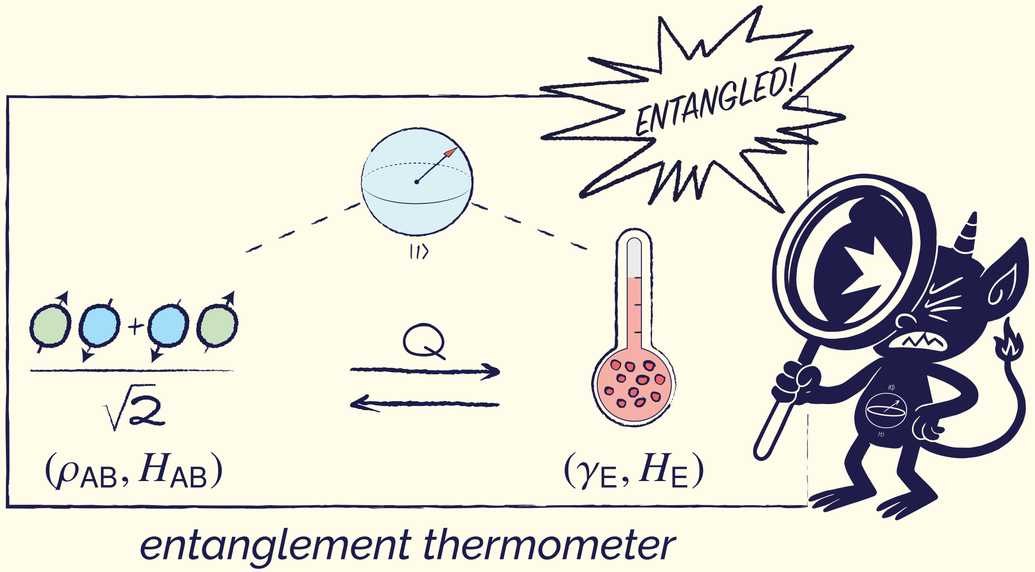
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Entanglement witness

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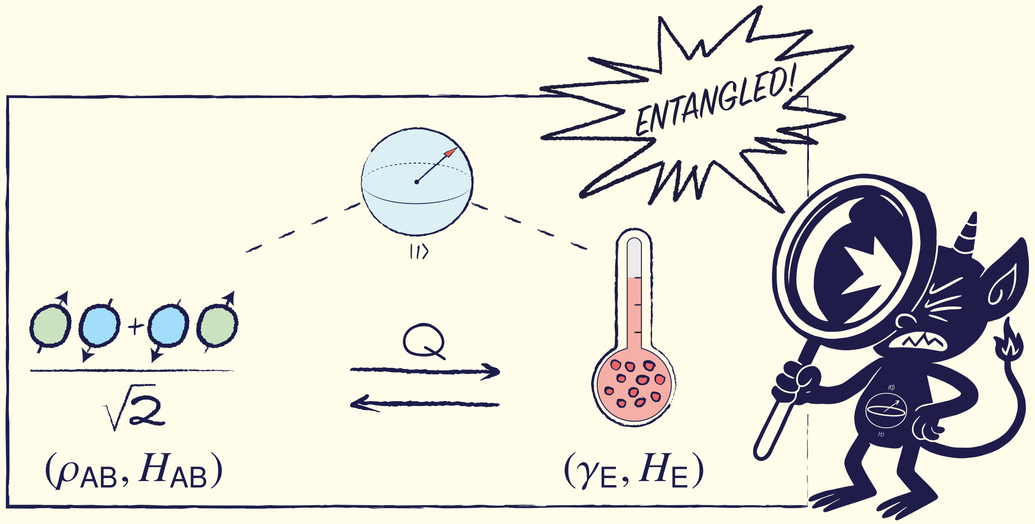


entanglement thermometer

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- **Isotropic state:** $\rho_\lambda = (1-\lambda)\left|\psi^+\rangle\langle\psi^+\right| + \lambda\frac{\mathbb{1}}{d^2}$, $H = \sum_{n=0}^{d-1} n\,|n\rangle\langle n|$

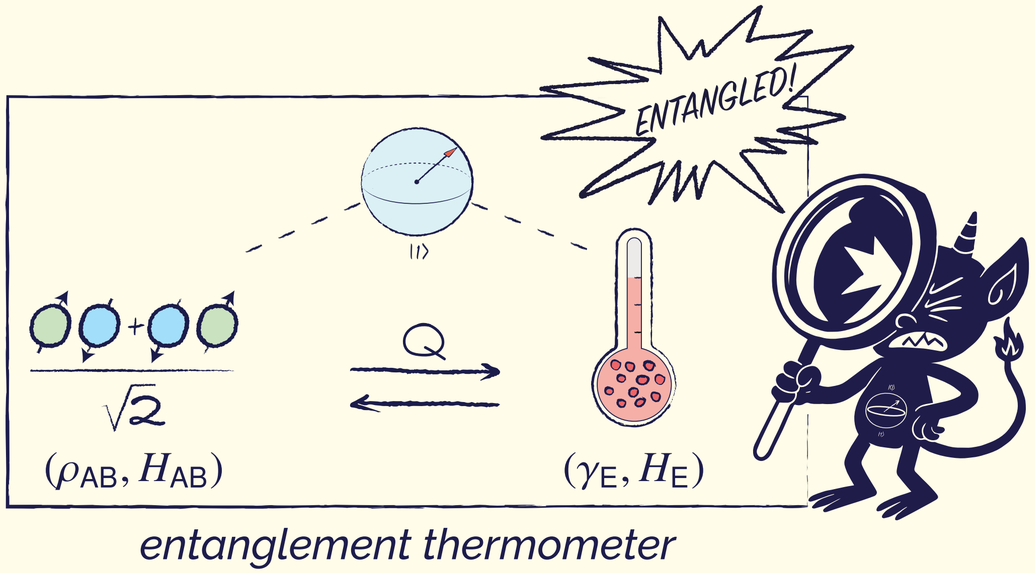
$$\left|\psi^+\right\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^{d}|ii\rangle$$

- **Enviroment @** $\beta = \frac{1}{2}$

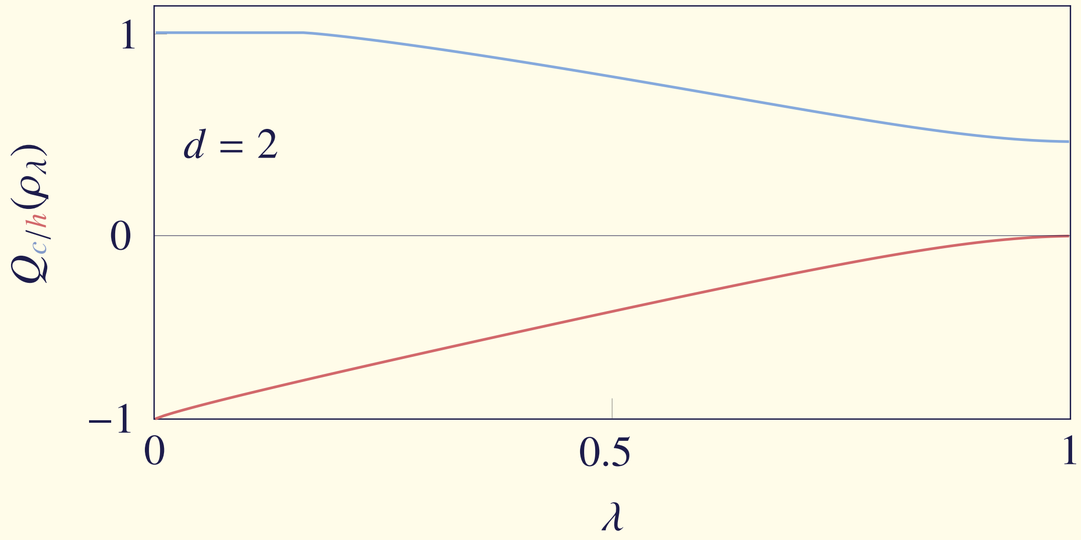
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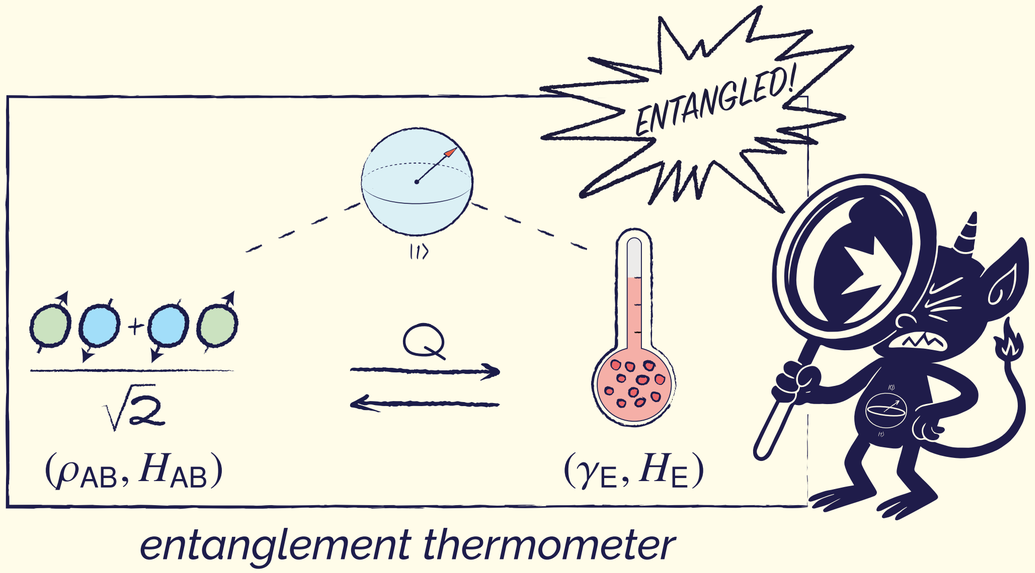
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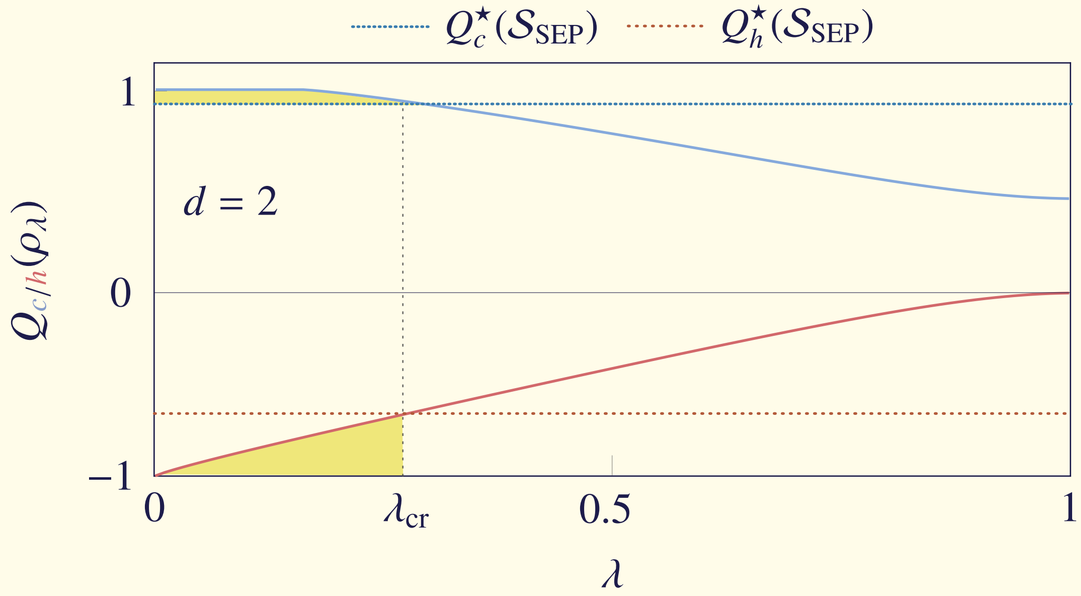
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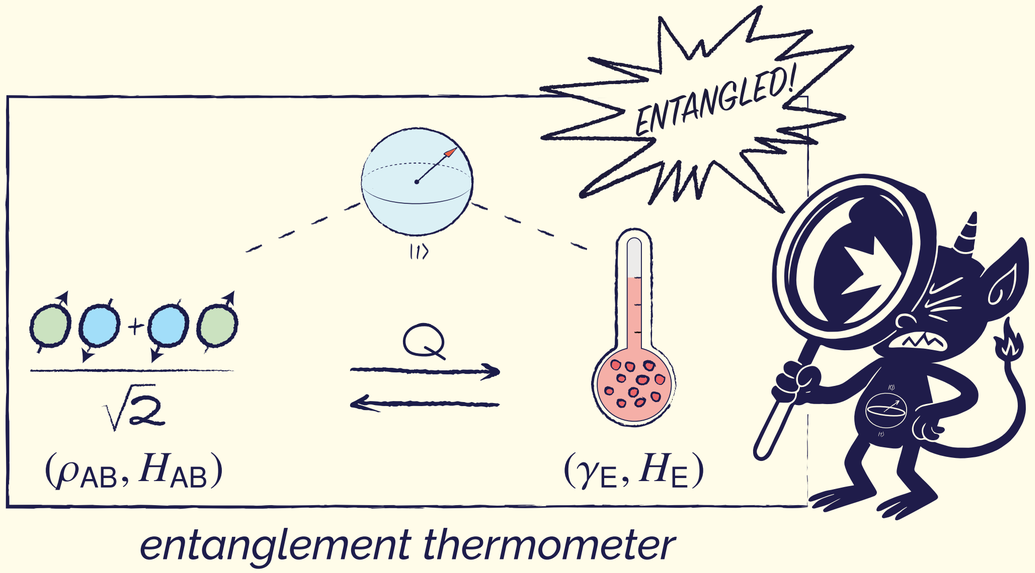
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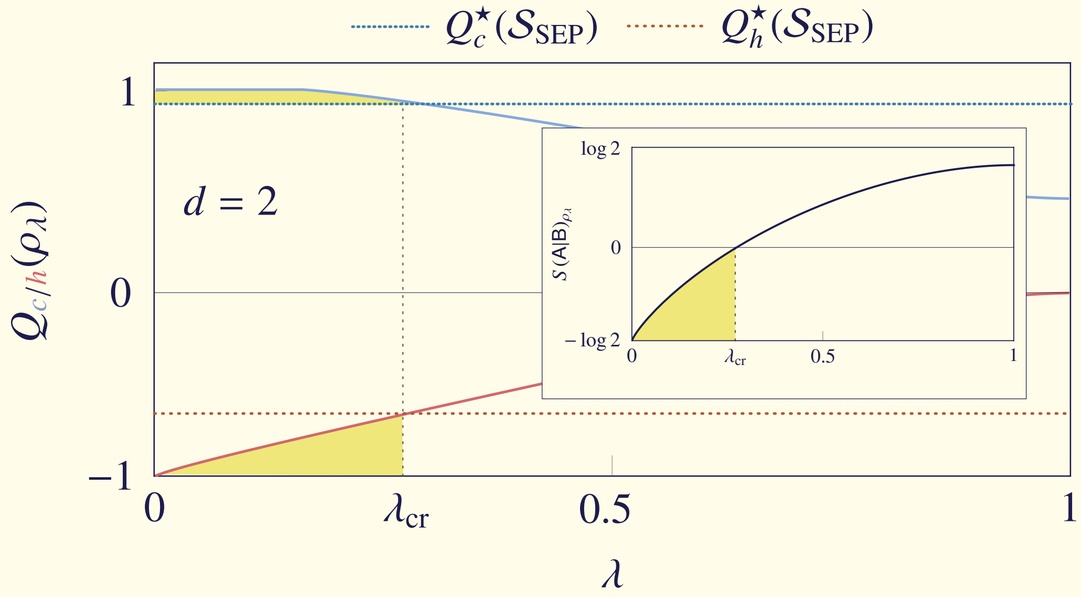


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$$\left|\psi^+\right\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^{d}|ii\rangle$$

- **Enviroment** @ $\beta = \frac{1}{2}$



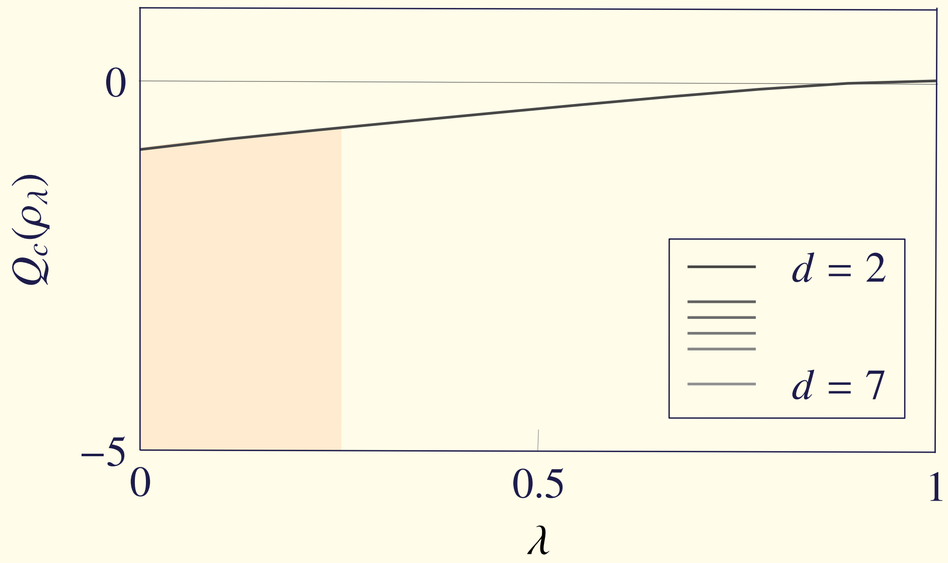
How about for higher d?

&

What's the limit of our detection?

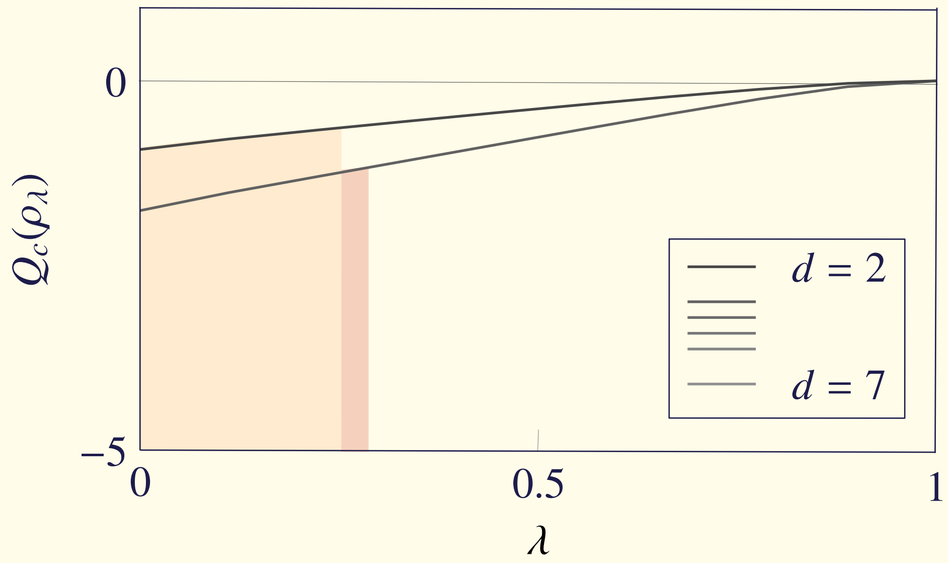
Entanglement witness

- Our detection improves with d:



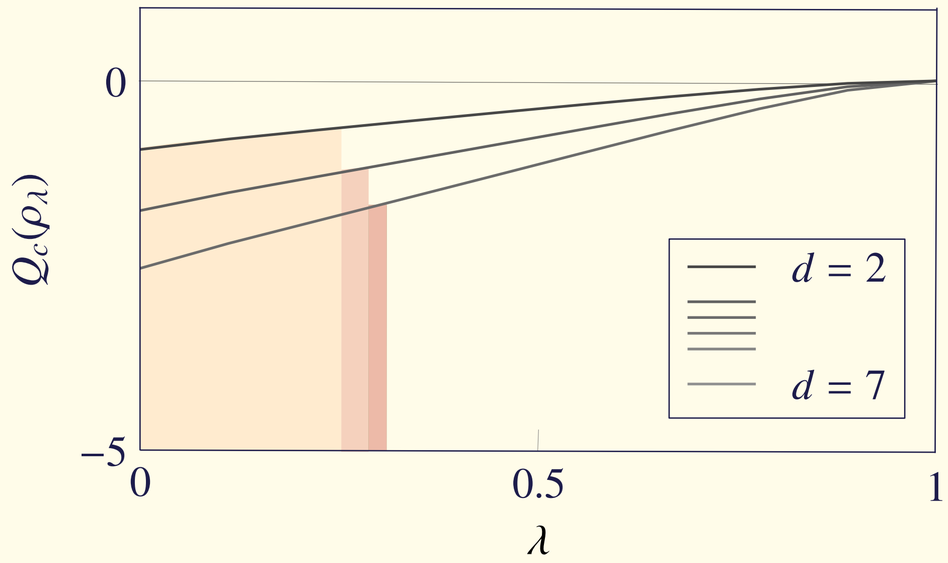
Entanglement witness

- Our detection improves with d:



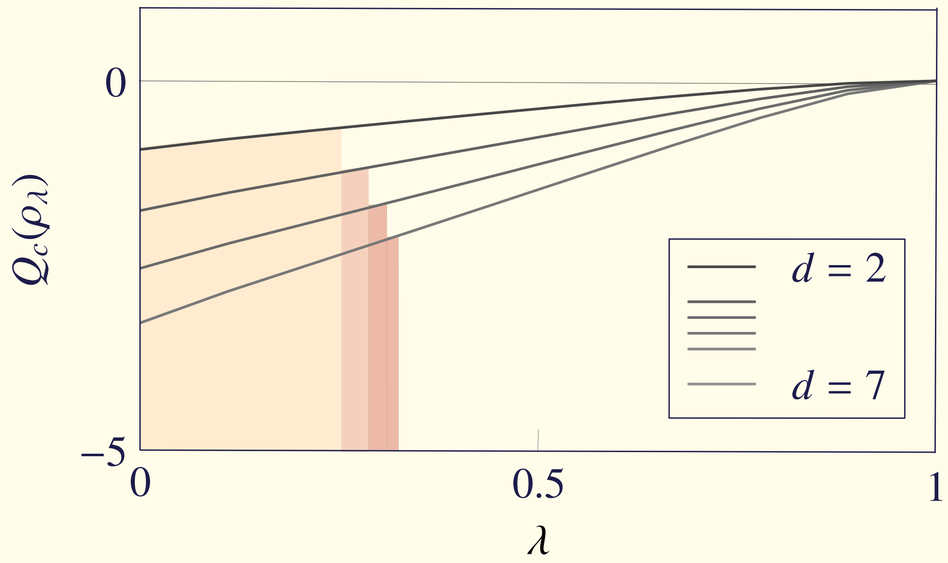
Entanglement witness

- Our detection improves with d:



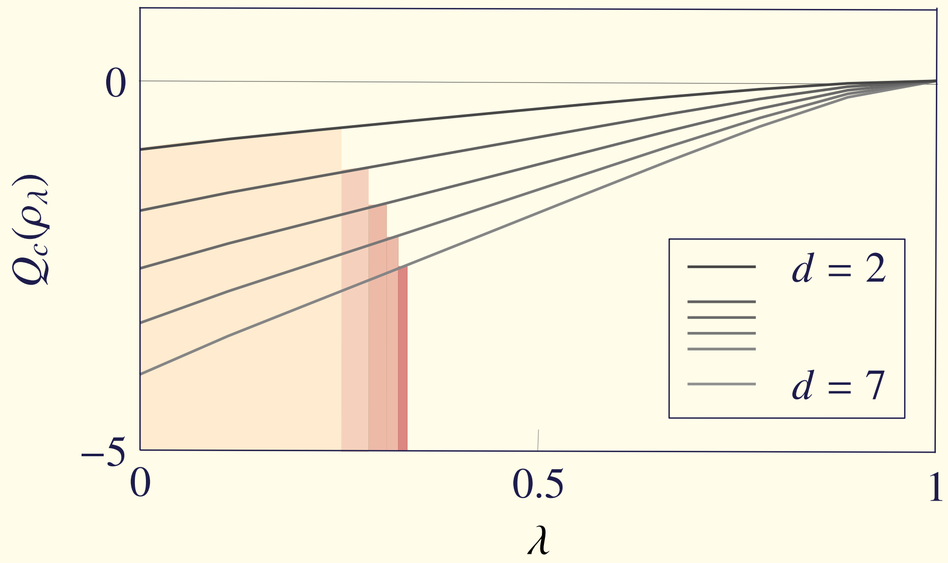
Entanglement witness

- Our detection improves with d:



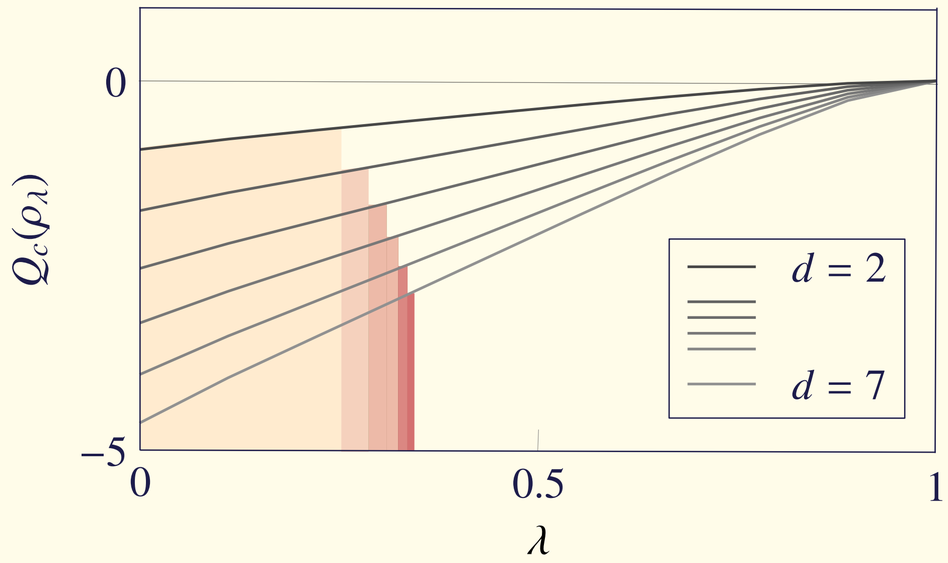
Entanglement witness

- Our detection improves with d:



Entanglement witness

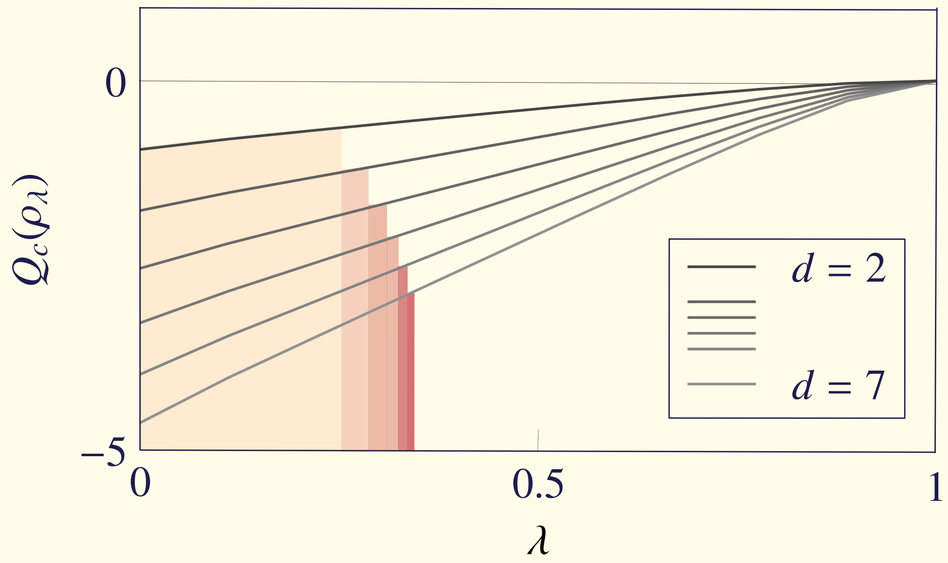
- Our detection improves with d:



A. de Oliveira Junior

Entanglement witness

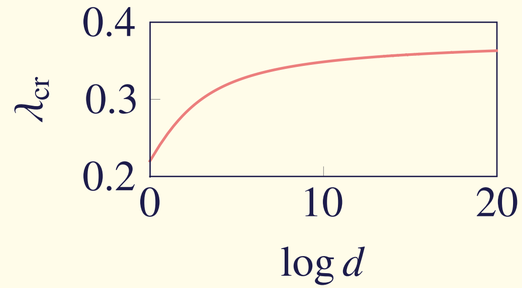
- Our detection improves with d:



- For arbitrarily dimensions our witness can detect up to

$$\left(\lambda_{\text{crt}} - d^2\lambda_{\text{crt}}\right)\log\left(\frac{\lambda_{\text{crt}}}{d^2}\right) + \left[d^2(\lambda_{\text{crt}} - 1) - \lambda_{\text{crt}}\right]\times \\ \times\log\left[\left(\frac{1}{d^2} - 1\right)\lambda_{\text{crt}} + 1\right] = d^2\log d$$

only looks ugly
(but can be easily solved)



Outlook

What I talked about

- **Thermodynamic process** as a new type of **entanglement witness**

What I haven't talked about
(but I wanted to):

- Witnessing **other** quantum properties:



- **Multipartite** quantum correlations?

